

Rubel Roy

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PROFILE SUMMARY

Aspiring Data Scientist with a strong foundation in Machine Learning, Deep Learning, and Data Analysis. Experienced in building and deploying end-to-end AI applications, with a passion for solving real-world problems through data-driven solutions.

EDUCATION

St. Xavier's College

Master of Science in Data Science

Kolkata, West Bengal, India

Expected Graduation: 2027

Gurudas College

Bachelor of Science in Physics

Kolkata, West Bengal, India

Graduated: 2024

PROJECTS

Real-Time Facial Expression Recognition using Deep Learning | [GitHub](#)

Ongoing

- Designed and trained a custom CNN from scratch on the FER2013 dataset (**35,887** grayscale facial images across **7** emotion classes) for facial expression recognition.
- Built an end-to-end computer vision pipeline including TensorFlow data loading, image augmentation, class-weighted training, and evaluation, achieving **52.76%** validation accuracy on unseen data.
- Engineered a CNN comprising 3 convolutional blocks with Batch Normalization, Dropout, and Global Average Pooling for efficient real-time inference.
- Performed detailed error analysis using confusion matrices, precision, recall, F1-score, and per-class accuracy, identifying Fear (11.62%) and Angry (27.14%) as the primary failure cases for future model improvements.
- Integrated OpenCV-based face detection with the trained CNN to perform real-time facial emotion prediction from webcam input, with deployment through a lightweight Flask web application in progress.
- Tech Stack:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, OpenCV

Collaborative Filtering-Based Book Recommendation Engine | [GitHub](#) | [Live Demo](#)

May 2026

- Developed an **item-based collaborative filtering** recommendation engine using the Book-Crossing dataset by constructing a user-item interaction matrix and computing **cosine similarity** between books.
- Preprocessed and filtered over **1 million+ user ratings** by removing inactive users and low-engagement books to improve recommendation quality and reduce sparsity.
- Implemented a Flask web application featuring a **Top-50 Popular Books** dashboard and real-time search-based recommendations with dynamically rendered book metadata and cover images.
- Built a low-latency inference pipeline by precomputing item similarity matrices and serializing model artifacts using Pickle, then deployed the application on Render.
- Tech Stack:** Pandas, NumPy, Scikit-learn, Flask, Jinja2, HTML, CSS, Bootstrap, Pickle, Render

TECHNICAL SKILLS

Programming Languages: Python, SQL (MySQL), R, HTML/CSS

Machine Learning: Scikit-learn, TensorFlow, Keras, XGBoost, Random Forest, CNNs, Collaborative Filtering, Cosine Similarity, Model Evaluation

Computer Vision: OpenCV, Image Preprocessing, Data Augmentation, Face Detection, Facial Expression Recognition

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Web Development: Flask, Jinja2, Bootstrap, Streamlit

MLOps & Tools: Git, GitHub, Docker, Jupyter Notebook, VS Code, Render

CERTIFICATIONS

Deep Learning Specialization – DeepLearning.AI (Andrew Ng)

[Credential](#)

Google Data Analytics Professional Certificate – Google (Coursera)

[Credential](#)

Unsupervised Learning in Python – DataCamp

[Credential](#)